Initializing the CMA-ES Covariance Matrix on the Simplex via a Fisher–Rao Orthonormal Frame

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Abstract

We optimize a function $H(x)$ over the scaled probability simplex $\Delta^R_{n-1} = \{x \in \mathbb{R}^n : x_i \geq 0,\ \sum_i x_i = R\}$ using a Riemannian variant of CMA-ES. The search distribution is carried in the tangent space of the simplex under the Fisher–Rao metric. This note records how to construct a g_m-orthonormal frame for that tangent space, how to initialize the covariance in it, and the design consequences of using rank-one and rank-μ covariance updates.

1 Tangent-space parameterization

We maintain a distribution mean $m \in \operatorname{int} \Delta^R_{n-1}$ and a step size $\sigma > 0$. The covariance is represented as an $(n-1) \times (n-1)$ positive-definite matrix C expressed in an orthonormal basis $\{e_1, \dots, e_{n-1}\}$ of the tangent space $T_m\Delta^R_{n-1}$ under the Fisher–Rao metric g_m.

The tangent space of the simplex at m is

$$T_m\Delta = \ker(\mathbf{1}^\top) = \Big\{v \in \mathbb{R}^n : \textstyle\sum_i v_i = 0\Big\}, \tag{1}$$

an $(n-1)$-dimensional subspace: any admissible displacement must preserve the sum constraint $\sum_i x_i = R$. A convenient frame is any orthonormal frame with respect to g_m; a systematic construction is to take a standard Euclidean orthonormal basis (ONB) of $\ker(\mathbf{1}^\top)$ and then g_m-orthonormalize it (e.g. via Gram–Schmidt, or a Cholesky factor of the metric tensor restricted to that basis).

Dimensional bookkeeping. The mean m has length n, but the covariance C, the standard-normal sampling vectors z, and the eigenfactors of C are all $(n-1)$-dimensional. The frame $E = [\, e_1 \cdots e_{n-1} \,]$ is an $n \times (n-1)$ matrix mapping tangent coordinates to ambient displacements.

2 The Fisher–Rao metric

On the simplex the Fisher–Rao metric is diagonal in the ambient coordinates,

$$g_m(u, v) = \sum_{i=1}^{n} \frac{u_i v_i}{m_i} \qquad \Longleftrightarrow \qquad G = \operatorname{diag}\left(\frac{1}{m_1}, \dots, \frac{1}{m_n}\right). \tag{2}$$

For the scaled simplex this coincides with the categorical Fisher metric up to a global constant $1/R$, which merely rescales σ. Note that G diverges as any $m_i \to 0$, so m must remain in the *interior* of the simplex.

3 Constructing the g_m-orthonormal frame

Step 1: Euclidean ONB of $\ker(\mathbf{1}^\top)$

Rather than a numerical QR factorization, use the closed-form *Helmert contrast matrix*, an $n \times (n-1)$ basis whose column k ($k = 1, \dots, n-1$) is

$$b_k = \frac{1}{\sqrt{k(k+1)}} \big(\underbrace{1, \dots, 1}_{k},\ -k,\ 0, \dots, 0\big)^\top. \tag{3}$$

Each column sums to zero (hence lies in $\ker(\mathbf{1}^\top)$), has unit Euclidean norm, and the columns are mutually orthogonal:

$$\mathbf{1}^\top b_k = 0, \qquad \|b_k\|_2 = 1, \qquad b_j^\top b_k = \delta_{jk}. \tag{4}$$

Step 2: restrict the metric and factor it

Let $B = [\, b_1 \cdots b_{n-1} \,]$. The metric restricted to the tangent basis is the $(n-1) \times (n-1)$ symmetric positive-definite matrix

$$M = B^\top G B, \tag{5}$$

with Cholesky factorization

$$M = LL^\top, \qquad L \text{ lower triangular.} \tag{6}$$

Step 3: orthonormalize

Define the frame

$$E = B\, L^{-\top}. \tag{7}$$

Then E is g_m-orthonormal:

$$E^\top G E = L^{-1} B^\top G B\, L^{-\top} = L^{-1} M L^{-\top} = L^{-1}(LL^\top)L^{-\top} = I_{n-1}. \tag{8}$$

The covariance is initialized to the identity in this frame, $C = I_{n-1}$, so that the ambient covariance operator is $ECE^\top$, a rank-$(n-1)$ operator supported on the tangent space.

Reference implementation (Eigen)

```
// Helmert: Euclidean ONB of ker(1^T), n x (n-1)
Mat helmert_basis(int n) {
    Mat B = Mat::Zero(n, n - 1);
    for (int k = 1; k <= n - 1; ++k) {
        double s = 1.0 / std::sqrt(double(k) * (k + 1));
        for (int i = 0; i < k; ++i) B(i, k - 1) = s;
        B(k, k - 1) = -double(k) * s;
    }
    return B;
}

// Build g_m-orthonormal tangent frame E (n x n-1) at interior mean m
void init_frame(const Vec& m) {
    Mat B = helmert_basis(n);
```

```
    Mat GB = m.cwiseInverse().asDiagonal() * B; // G*B: scales row i by 1/m_i
    Mat M = B.transpose() * GB; // restricted metric (n-1 x n-1)
    Eigen::LLT<Mat> llt(M);
    if (llt.info() != Eigen::Success)
        throw std::runtime_error("restricted metric not SPD (m on boundary?)");
    Mat L = llt.matrixL();
    // E = B L^{-T} <=> E^T = L^{-1} B^T (lower-triangular solve)
    E = L.triangularView<Eigen::Lower>().solve(B.transpose()).transpose();
    C = Mat::Identity(n - 1, n - 1); // covariance in this frame
}
```

4 Practical notes

- **Initialize at the centroid.** Take $m_i = R/n$. This is the natural interior start and is well-conditioned: there $G = (n/R)I$, so $M = (n/R)I$ and $E = \sqrt{R/n}\,B$.
- **Exploit diagonality.** G is diagonal; never materialize it densely. Scaling the rows of B by $1/m_i$ computes GB directly.
- **Retraction (later).** A sampled tangent vector $v = \sigma\, E(B_C D_C z)$ is mapped back to the simplex by the exponential map $m \mapsto \mathrm{Exp}_m(v)$. The isometry $x \mapsto 2\sqrt{x}$ sends the simplex to a sphere, so Fisher–Rao geodesics are great-circle arcs.

5 Verification

After `init_frame` at an interior mean, check numerically:

(1) $\|E^\top G E - I_{n-1}\| < 10^{-10}$ $\quad$ (g_m-orthonormality);

(2) $\mathbf{1}^\top E \approx 0$ $\quad$ (every column is a valid tangent vector);

(3) at the centroid, $E^\top E = (R/n)\, I_{n-1}$.

6 Consequence of rank-one and rank-μ updates

The covariance is updated each generation by a combination of a rank-one and a rank-μ term,

$$C \leftarrow (1 - c_1 - c_\mu)\, C \;+\; c_1\, p_c p_c^\top \;+\; c_\mu \sum_i w_i\, y_i y_i^\top, \tag{9}$$

where y_i are the selected tangent steps and p_c is the covariance evolution path. The rank-μ term is built entirely from the current generation's data and is self-contained in the current frame. The rank-one term, however, uses the *cumulative* evolution path

$$p_c \leftarrow (1 - c_c)\, p_c + \sqrt{c_c(2 - c_c)\,\mu_{\mathrm{eff}}}\;\langle y\rangle_w, \tag{10}$$

which carries history across generations, as does the step-size path p_σ used by cumulative step-length adaptation.

Why this matters. The frame $E = E(m)$ is mean-dependent and changes every generation. The evolution paths p_c, p_σ and the covariance C are stored in tangent *coordinates*, so when the mean moves $m_t \to m_{t+1}$ these objects live in different tangent spaces and cannot be combined directly. Whereas a pure rank-μ scheme uses only current-generation data and is forgiving, the rank-one and CSA paths make **parallel transport mandatory**.

Parallel transport. Moving the mean along the Fisher–Rao geodesic induces a linear isometry $\tau : T_{m_t} \to T_{m_{t+1}}$. In the two g-orthonormal frames it is represented by the orthogonal matrix

$$Q = E_{t+1}^\top \, G_{m_{t+1}} \, \tau \, E_t, \qquad Q^\top Q = I_{n-1}, \tag{11}$$

and the persistent state is transported as a batch each generation,

$$p_c \leftarrow Q p_c, \qquad p_\sigma \leftarrow Q p_\sigma, \qquad C \leftarrow Q \, C \, Q^\top. \tag{12}$$

Because Q is orthogonal, $QCQ^\top$ remains positive-definite with the same eigenvalues, and path norms are preserved—exactly the property that keeps the CSA length comparison $\|p_\sigma\|$ versus its expectation valid.

Resulting loop structure.

1. Sample λ offspring in T_{m_t};
2. select and recombine to update the mean $m_t \to m_{t+1}$;
3. build the new frame $E_{t+1} = E(m_{t+1})$;
4. transport p_c, p_σ, C from the old frame to the new via Q;
5. apply the CSA update (p_σ, σ) and the covariance update (9)–(10);
6. eigendecompose $C = B_C D_C^2 B_C^\top$ for the next round of sampling.

The frame *construction* itself is unchanged by the choice of update; only the surrounding state and loop must account for transport.